DEPARTMENT OF CHEMICAL ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY MADRAS
CHENNAI – 600036

Hybrid Data Augmentation Techniques for Disease Identification from Clinical Notes

A Project Report

Submitted by

KAUSTABH GANGULY

For the award of the degree

Of

MASTER OF TECHNOLOGY

February 2025

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"Artificial intelligence will not replace the physician, but it will empower them to deliver more personalized care by freeing them from administrative burdens." — ***Eric Topol***

This work is dedicated to the incredible community of teachers, mentors, and classmates who
helped shape my understanding and curiosity. Your wisdom, shared enthusiasm,
and kind guidance have propelled me through every challenge.

Special thanks to my professors, who brought complex ideas to life, and to my mentors, whose faith in my potential and real-world insights
enriched my knowledge in ways a classroom alone could never provide.

To all of you, I am deeply grateful.

THESIS CERTIFICATE

This is to undertake that the Project Report titled **HYBRID DATA AUGMENTATION TECHNIQUES FOR DISEASE IDENTIFICATION FROM CLINICAL NOTES**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master of Technology**, is a bona fide record of the research work done by me under the supervision of **Mr. Samyabrata Chakraborty (Lead Solution Architect & Head of Centre of Excellence, TCS), Mr. Debopam Nanda (Lead Data Scientist, TCS)** .

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my mentors, Mr. Samyabrata Chakraborty and Mr. Debopam Nanda, for their invaluable guidance and expertise in clinical natural language processing and data science. I am also deeply grateful to my professors at the Indian Institute of Technology Madras for generously sharing their knowledge and support. My heartfelt thanks go to Dr. Alka Bhushan for her thoughtful guidance on report writing and constructive feedback, which notably enhanced this work. Lastly, I am indebted to my family and friends for their unfailing encouragement and belief in my efforts throughout this journey.

ABSTRACT

Automated ICD coding has improved substantially in recent years, yet many systems struggle with rare disease codes because of limited training data and the high granularity of ICD-10-CM. This thesis addresses that shortfall by combining synthetic clinical text generation, knowledge-driven validation, and memory-efficient model training to enhance both accuracy and coverage for underrepresented codes. Large language models (LLMs) are employed to produce factually diverse discharge summaries, guided by ontologies such as SNOMED CT and Orphanet. A stringent multi-phase review pipeline—encompassing rule-based checks, LLM-based fact verification, and ICD feedback loops—filters out unrealistic samples and preserves label integrity. Subsequently, these validated synthetic records augment existing training datasets for multi-label classification models, including transformer-based architectures (e.g., PLM-ICD) and synonym-aware networks. Emphasis is placed on boosting recall for rare codes, where conventional methods typically falter. Improvements in macro-F1 are observed once data scarcity is alleviated through synthetic augmentation. In addition, practical constraints are addressed by integrating approaches such as gradient checkpointing, quantization (QLoRA), and knowledge distillation, enabling resource-limited systems to fine-tune sizable language models without compromising precision. Beyond performance metrics, the thesis underscores the importance of model transparency and trustworthiness. Techniques like attention visualization and SHAP-based attributions highlight key textual fragments responsible for each predicted label, offering insights that can support clinical validation and audit processes. Error analyses further uncover any systematic biases, guiding refinements to both the data generation and classification pipelines. By merging robust augmentation, structured knowledge, and carefully optimized training routines, this work advances a scalable, interpretable ICD coding framework that narrows gaps in rare disease coverage and paves the way for more comprehensive clinical AI solutions.

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NOTATION

α	A learning rate or weight factor in optimization algorithms
$\mathcal{L}$	The complete label space of ICD codes
BCE	Binary Cross Entropy, a common loss function for multi-label learning
F1	Harmonic mean of Precision and Recall, used for model evaluation
θ	Trainable parameters of the classification model
$\tilde{x}_j$	Synthetic clinical note generated for augmentation
D	Overall dataset of paired clinical notes and labels
$f(\cdot)$	The learned multi-label classification function mapping text to sets of labels
$P(Y \mid x; \theta)$	Predicted probability of codes Y given note x
x_i	The i-th clinical note (free-text discharge summary)
Y_i	The set of ICD codes associated with the i-th clinical note

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO THIS THESIS

The *International Classification of Diseases (ICD)* is a globally recognized coding system maintained by the World Health Organization (Organization, 2019), with its most recent iterations being ICD-10 and ICD-11. This system was developed to standardize the labeling of diseases, medical procedures, and underlying health conditions, thereby fostering interoperability across diverse healthcare environments. As a result, patient diagnoses and treatments are recorded more consistently, allowing healthcare providers to communicate more effectively, researchers to examine data with fewer ambiguities, and public health authorities to craft evidence-based policies.

When multiple ICD codes are assigned to a single patient's record, the information becomes richer by including primary diagnoses and related comorbidities. Large-scale databases, such as the *Medical Information Mart for Intensive Care-IV (MIMIC-IV)*, reflect this practice by covering not only the chief complaint but also secondary or rare conditions (Johnson *et al.*, 2023). A major concern, however, arises if any of these codes are misassigned. Inaccuracies and omissions can ripple through subsequent clinical decision-making, research conclusions, and billing outcomes. Hence, the role of accurate coding becomes paramount in bridging the gap between raw clinical documentation and reliable medical insights.

From a financial standpoint, healthcare institutions depend on correct coding to claim reimbursements, and errors—either over-coding or under-coding—risk financial imbalances. Studies have highlighted the direct impact of clinical documentation on

billing accuracy and institutional revenue, stressing the importance of precision in ICD assignments (Dong *et al.*, 2022). At a population level, aggregated ICD-coded data forms the bedrock of epidemiological tracking, policy development, and larger-scale medical research. Erroneous codes can alter the trajectory of these studies, potentially skewing public health interventions (Edin *et al.*, 2023).

Nevertheless, the manual process of ICD coding remains both tedious and vulnerable to human error. Coders routinely parse extensive and often unstructured discharge summaries that include technical jargon, acronyms, or shorthand specific to a given clinical context (Wrenn *et al.*, 2010). Rare or lesser-known conditions risk being misrepresented or overlooked in the daily rush of high-volume coding workloads. Automated or semi-automated tools have therefore been suggested as an avenue for alleviating these problems. Such solutions provide preliminary code assignments that human coders can validate, thus streamlining workflow and improving data quality (Campbell and Giadresco, 2020).

Over the years, numerous computational methods have been devised to assist in the assignment of ICD codes. Early rule-based systems depended on dictionaries and handcrafted rules (Farkas and Szarvas, 2008), but these were limited in scalability and flexibility. Traditional machine learning techniques (e.g., support vector machines and logistic regression) made modest progress but often stumbled over the diverse, nuanced terminology of clinical text (Scheurwegs *et al.*, 2017; Perotte *et al.*, 2014). Momentum grew significantly with the advent of deep learning, starting with convolutional neural networks (CNNs) (Kim, 2014; Mullenbach *et al.*, 2018) and recurrent neural networks (e.g., LSTMs (Hochreiter and Schmidhuber, 1997)) that learned hierarchical or temporal patterns from unstructured notes. More recently, transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) have improved performance in many language tasks (Devlin *et al.*, 2019), though they often struggle

with extremely long documents and heavily imbalanced data (Gao *et al.*, 2021; Huang *et al.*, 2022).

Several specialized strategies have since been proposed to address these challenges, including extended context transformers, hierarchical embeddings, and curriculum learning over the ICD taxonomy (Nickel and Kiela, 2017; Ren *et al.*, 2022; Bengio *et al.*, 2009). These refinements aim to capture relevant details in lengthy clinical narratives and to handle the long-tail distribution in which a few codes dominate while a majority of rare codes receive minimal coverage (Rios and Kavuluru, 2018). Alongside these methodological expansions, interpretability is receiving growing attention in automated ICD coding: clinicians must verify or scrutinize a model's reasoning to ensure that each assigned code is justified by actual clinical evidence (Holzinger *et al.*, 2017; Warner *et al.*, 2024).

In parallel, new lines of research have begun focusing on *entity recognition* and *ontology-based approaches* to enhance the robustness of coding systems. Tools like *cTAKES* (Savova *et al.*, 2010) and *MedCAT* (Kraljevic *et al.*, 2021) identify medical concepts in text and map them to standardized vocabularies, such as SNOMED CT (Donnelly, 2006) or the Unified Medical Language System (UMLS) (Bodenreider, 2004). Although these pipelines do not automatically produce final ICD codes, they offer structured terminological insights that can reduce ambiguity in free-text data (Wu *et al.*, 2018). More advanced systems incorporate ICD-specific knowledge directly into the classification process (e.g., hyperbolic embeddings of the ICD hierarchy (Cao *et al.*, 2020) or label descriptions drawn from official manuals). This knowledge-aware approach has consistently shown promise in bridging the gap between unstructured narratives and the highly specialized code sets used in clinical practice.

An especially difficult hurdle is how to improve the recognition of rare or "zebra" codes. When diagnoses occur infrequently, data-driven models often lack sufficient positive examples for generalization (Rios and Kavuluru, 2018). To combat this scarcity, researchers have begun to explore *synthetic data generation*. By leveraging large language models (LLMs) to create artificial discharge summaries or augment existing text, the model's exposure to underrepresented conditions can be increased (Falis *et al.*, 2022, 2024). Ontologies like SNOMED CT or the UMLS help ground these generated examples so that the synthetic narratives remain medically plausible (Kumichev *et al.*, 2024). For instance, a rare genetic disorder might have very few real cases in a training corpus; generating extra samples keyed to that condition's known symptomatology can yield substantial performance gains, particularly in recall for low-frequency labels.

These generative methods build on a broader trend in natural language processing, in which powerful transformers (e.g., GPT-3.5 or GPT-4) are used to produce text that mimics human writing. Indeed, recent work suggests that LLMs prompted with curated medical knowledge can synthesize relatively realistic patient histories (Yang *et al.*, 2023). Although concerns persist about potential hallucinations or a lack of full clinical richness, experiments have shown that carefully designed prompts—possibly incorporating knowledge graphs or code descriptions—can yield synthetic notes beneficial for model training. Complementary approaches use synonyms or sibling diagnoses from an ontology to systematically "perturb" or augment existing text (Yuan *et al.*, 2022), thus expanding the model's exposure to variant language for the same condition.

However, practical integration of these advanced methods requires consideration of real-world constraints. Data privacy regulations often limit the availability of large, richly annotated clinical corpora. Synthetic notes, if generated responsibly, can alleviate privacy concerns while still reflecting key features of real patient reports. Yet computational cost and resource constraints remain a challenge: training or fine-tuning long-context

transformers can be expensive, and using LLM APIs extensively may be prohibitively costly for some healthcare institutions (Ji *et al.*, 2021). Techniques such as quantization, gradient checkpointing, or hierarchical batching (Geiping and Goldstein, 2023) can reduce resource demands. Another paramount concern is reliability: automated coding systems must provide transparent justifications for the codes they produce if they are to earn the trust of clinicians. Interpretable mechanisms, such as label-specific attention or post-hoc attribution methods (e.g., Integrated Gradients, SHAP), facilitate manual review and verification (Teng *et al.*, 2020).

To address these persistent challenges, this thesis proposes a framework that combines ontology-informed data augmentation, cutting-edge neural architectures, and interpretability techniques to enhance automated ICD coding. In particular, four major strategies anchor the work:

- **Synthetic Data Generation for Rare Codes:** Large language models are leveraged in tandem with medical ontologies (e.g., SNOMED CT, UMLS) to produce realistic training examples for less frequently occurring conditions (Kumichev *et al.*, 2024; Falis *et al.*, 2022). By enriching the dataset with synthetic discharge summaries, we strive to close the performance gap between common and rare codes.

- **Knowledge-Aware Multi-Label Architectures:** Neural models capable of handling thousands of code labels are examined, including transformer-based classifiers with extended input contexts. Techniques such as hyperbolic embeddings of the ICD hierarchy (Nickel and Kiela, 2017; Cao *et al.*, 2020) and label attention mechanisms (Vu *et al.*, 2020*b*) are integrated to improve performance and interpretability in multi-label classification.

- **Clinically Oriented Interpretability:** Interpretability remains crucial for adoption in clinical workflows. Methods like SHapley Additive exPlanations (SHAP), attention heatmaps, or concept-level attributions (Holzinger *et al.*, 2017; Teng *et al.*, 2020) are incorporated to help coders and physicians quickly validate or question the system's logic, especially for critical or rare diagnoses.

- **Resource-Conscious Training and Deployment:** Training strategies such as quantization and gradient checkpointing are used to mitigate heavy computational requirements (Geiping and Goldstein, 2023). This ensures that our proposed system remains accessible to healthcare organizations with limited GPU resources.

By uniting ontology-driven augmentation, state-of-the-art neural methods, and transparent validation, this work aims to significantly improve both the breadth and reliability of ICD coding systems. In so doing, we anticipate that more accurate, explainable coding will foster better clinical documentation, refine large-scale epidemiological analyses, and ultimately contribute to more effective patient care and public health interventions. The chapters that follow delve into the technical foundations, present empirical results, and discuss the real-world feasibility of deploying such a system in modern healthcare settings.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION AND THE EVOLUTION OF ICD CODING

The automation of the International Classification of Diseases (ICD) coding process has long been recognized as a critical area of research in medical informatics and Natural Language Processing (NLP). The ICD system, maintained by the World Health Organization, is used worldwide to classify diseases and health conditions for epidemiological research, healthcare management, and clinical billing. Accurate and efficient coding is essential because ICD codes capture details of a patient's diagnosis, comorbidities, and complications, thereby supporting reliable record-keeping and reimbursement processes (Organization, 2019).

Originally, the Ninth Revision (ICD-9) provided a limited number of disease categories. Over time, ICD-9 was phased out in favor of ICD-10, driven by the need for more specificity and additional capacity to accommodate new diseases (Dong *et al.*, 2022). ICD-10-CM (Clinical Modification) expanded the code set substantially (from approximately 13,000 to 68,000 codes) and introduced alphanumeric codes capable of encoding disease severity, anatomical site, and laterality (Edin *et al.*, 2023). Although this granularity supports more detailed patient records, it also increases complexity in manual coding workflows (Farkas and Szarvas, 2008).

The progression to ICD-11 (Organization, 2019) further reflects international efforts to enhance coding flexibility and align medical concepts more precisely with emerging public health priorities. Its hierarchical structure and additional extensions aim to capture nuances of disease presentation, enabling more robust epidemiological tracking. However, these regular revisions amplify the complexity for human coders, who must keep abreast

of ever-expanding vocabularies and specialized rules (Scheurwegs *et al.*, 2017). The high stakes of accurate billing and statistical reporting underscore the importance of developing automated coding solutions that can function reliably, even when new codes and terminologies are introduced.

This chapter reviews the broad spectrum of approaches and challenges that characterize automated ICD coding. Early rule-based and machine learning (ML) methods are summarized, followed by deep learning approaches that leverage convolutional, recurrent, and transformer-based architectures. Recent improvements in long-context transformer designs, modern BERT-like encoders, knowledge-based methods, and curriculum learning strategies are discussed, highlighting progress in handling the extensive and imbalanced ICD label space. The chapter also explores data imbalance and rare-code prediction, synthetic data generation for underrepresented diagnoses, interpretability, and the practical challenges of deploying automated ICD coding systems in real-world clinical environments. Relevant evaluation metrics and future research directions are presented to offer a holistic overview of current capabilities and ongoing developments in this domain.

2.2 EARLY APPROACHES TO AUTOMATED ICD CODING

2.2.1 Rule-Based Systems and Traditional Machine Learning

Early efforts to automate ICD coding primarily involved rule-based systems and traditional machine learning techniques. Rule-based systems utilized handcrafted linguistic rules, domain-specific ontologies, and keyword matching heuristics to assign ICD codes to clinical narratives (Farkas and Szarvas, 2008; Scheurwegs *et al.*, 2017). These systems often achieved high precision under narrowly defined contexts because medical experts designed rules explicitly targeting common or well-understood diagnoses. Nonetheless, maintaining rule-based systems for extensive code sets was difficult; an entire rule set would often need re-engineering whenever revisions to ICD or local documentation

practices emerged (Scheurwegs *et al.*, 2017).

Parallel to rule-based approaches, researchers applied supervised classifiers such as logistic regression, Decision Trees, or Support Vector Machines (SVMs) trained on bag-of-words or TF-IDF representations of clinical notes (Perotte *et al.*, 2014). Although these methods sometimes benefited from straightforward interpretability (e.g., top weighted words for each code), they struggled with the extremely large and imbalanced label space, as ICD-9-CM and ICD-10-CM collectively encompass thousands of possible codes. Traditional feature engineering often could not capture the subtle semantic patterns indicative of specific diagnoses, particularly when these patterns were distributed across long or unstructured notes. Moreover, such approaches did not perform well on rare codes, which had scant training examples (Duque *et al.*, 2021). Despite these limitations, these early systems established important baselines and indicated that integrating domain knowledge (e.g., simple dictionaries of medical terms) could enhance precision.

2.2.2 Limitations and Motivation for Deep Learning

Although traditional ML solutions revealed the potential of automated ICD coding, they were often constrained by high-dimensional data, sparse features, and a lack of capacity to fully capture clinical context. The low coverage of rare codes and the complex textual nature of clinical documentation prompted researchers to investigate more robust methods that did not rely heavily on labor-intensive rule sets or shallow textual features. The subsequent shift toward deep learning was motivated by these challenges, as neural architectures offered new pathways for automatically learning hierarchical textual representations with minimal handcrafting (LeCun *et al.*, 2015).

2.3 DEEP LEARNING FOR AUTOMATED ICD CODING

2.3.1 Convolutional and Recurrent Neural Network Approaches

Deep learning opened up new avenues for capturing semantic and syntactic nuances in clinical text. Convolutional Neural Networks (CNNs) were among the earliest

neural architectures to be tested, owing to their effectiveness in extracting local n-gram features (Kim, 2014). In the context of ICD coding, Mullenbach et al. (Mullenbach *et al.*, 2018) introduced the Convolutional Attention for Multi-Label classification (CAML) model. CAML employed a convolutional encoder to represent textual segments, followed by a label-specific attention mechanism that highlighted the most relevant tokens for each code prediction. The increased interpretability of this attention mechanism was particularly appealing for healthcare applications, where transparency is critical:

$$\alpha^{(l)} = \text{softmax}(W^{(l)} h + b^{(l)}), \tag{2.1}$$

where h denotes the hidden representations from the convolutional layers, and $W^{(l)}$, $b^{(l)}$ are label-specific parameters.

Subsequent CNN-based models explored deeper architectures with residual connections and multi-filter designs. For instance, Li and Yu (Li and Yu, 2020) proposed the Multi-Filter Residual Convolutional Neural Network (MultiResCNN), which combines different kernel sizes and residual pathways to capture diverse syntactic and semantic patterns in clinical text. These refinements demonstrated measurable gains in F1-scores on benchmarks like MIMIC-III and other hospital datasets.

Recurrent Neural Networks (RNNs) offered an alternative strategy by encoding sequences of tokens in a step-by-step manner. Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) and Gated Recurrent Units (GRUs) were particularly suited for capturing longer-range dependencies in clinical notes. Vu et al. (Vu *et al.*, 2020*a*,*b*) employed a bidirectional LSTM with a label attention mechanism, where each label learned distinct attention weights over the text. This enabled more targeted coding decisions, accounting for the context needed by each diagnosis or procedure label. Although RNN models captured sequential information better than many CNN designs, they remained prone to vanishing gradients on very long

documents and could be computationally expensive.

By the late 2010s, CNN- and RNN-based models had markedly outperformed traditional ML baselines (Perotte *et al.*, 2014), demonstrating that neural networks could automatically learn powerful text features. Nevertheless, these architectures often required careful truncation or sampling of long notes, and they still struggled under conditions of data imbalance or when handling thousands of rarely seen ICD codes (Wrenn *et al.*, 2010).

2.4 TRANSFORMER-BASED ARCHITECTURES AND EXTENSIONS

2.4.1 Standard Transformers and Clinical Adaptations

Transformer-based models, especially BERT (Devlin *et al.*, 2019), revolutionized NLP by replacing the sequential encoding of RNNs with self-attention over entire sequences. This design facilitated more direct capture of long-range dependencies. However, standard BERT has a typical input limit of 512 tokens, which poses a challenge in clinical coding, where notes can span thousands of tokens (Gao *et al.*, 2021). Early attempts to integrate BERT into ICD coding often truncated notes, losing critical data.

Huang et al. (Huang *et al.*, 2022) introduced the PLM-ICD model, which segments each discharge summary into manageable chunks and processes them with a pre-trained BERT. A label attention mechanism then aggregates chunk representations that are relevant to specific codes. Although such approaches allowed BERT-based solutions to outperform many CNN and RNN baselines on micro-F1 scores, truncated documents risked overlooking important evidence for certain diagnoses (Dong *et al.*, 2022).

Heo et al. (Heo *et al.*, 2022) addressed this long-text issue by proposing the Document-to-Sequence BERT (D2SBERT) framework. It divides the full document into multiple fixed-length segments and processes each segment independently through BERT. A sequence-level attention mechanism then merges these representations, capturing essential

information across the entire document. This method outperformed various CNN-based benchmarks on MIMIC-III, demonstrating that transformer-based models can effectively handle extensive notes through segmenting and hierarchical attention.

2.4.2 Modern BERT-Based Encoders for ICD Coding

Researchers also investigated architectural modifications and domain-specific pretraining to accommodate larger input lengths, improve inference speeds, and address the domain-specific vocabularies of clinical notes. Several recent works introduce novel BERT-like encoders:

- **MosaicBERT** (Portes *et al.*, 2023) and **CrammingBERT** (Geiping and Goldstein, 2023): Optimize training efficiency, allowing them to achieve comparable BERT performance with shorter training times. They do not fundamentally resolve the difficulties associated with extremely long narrative texts but reduce overall resource demands.

- **NomicBERT** (Nussbaum *et al.*, 2024) and **GTE-en-MLM** (Zhang *et al.*, 2024): Extend the maximum input length from 2k to 8k tokens and introduce advanced unpadding methods. These models exhibit modest performance improvements on retrieval tasks and partially address memory constraints by employing sparser attention patterns.

- **ModernBERT** (Warner *et al.*, 2024): Incorporates architectural innovations such as alternating local-global attention, Rotary Positional Embeddings (RoPE), and label-aware attention modules. ModernBERT consistently outperforms prior baselines on both short and long ICD coding tasks, while reducing inference memory usage up to 8k tokens. Its hardware-aware design exemplifies how targeted architectural choices can mitigate computational overhead.

Despite progress, BERT-like transformers continue to encounter difficulties with extreme class imbalance, especially for codes encountered infrequently. Hardware constraints remain challenging when notes exceed thousands of tokens. Nonetheless, these tailored encoder designs underscore that carefully balancing domain knowledge, input length, and memory efficiency can produce state-of-the-art clinical coding models.

2.5 CURRICULUM LEARNING AND HIERARCHICAL KNOWLEDGE INTEGRATION

The hierarchical structure of ICD codes naturally suggests a curriculum learning approach, where models are initially exposed to broad categories before proceeding to more granular ones. Bengio et al. (Bengio *et al.*, 2009) formally introduced curriculum learning as a strategy to improve model performance by scheduling examples from simple to difficult.

Ren et al. (Ren *et al.*, 2022) applied a hierarchical curriculum (HiCu) by organizing ICD codes into tree-like levels. During training, they gradually refined predictions from higher-level ICD groupings (e.g., "diseases of the circulatory system") down to more specific subcodes. A hyperbolic embedding mechanism (Nickel and Kiela, 2017) transferred knowledge from parent-level predictions to child-level nodes, enabling the model to leverage relationships among ICD codes. This approach mitigated some aspects of data imbalance, demonstrated by improved macro-F1 and macro-AUC scores, particularly for infrequent codes. Consequently, curriculum learning has emerged as a complementary strategy to neural architectures, aligning naturally with the structured taxonomies in ICD-9-CM, ICD-10-CM, and ICD-11.

2.6 CHALLENGES IN AUTOMATED ICD CODING

2.6.1 Data Imbalance and Rare Codes

Automated ICD coding is inherently an extreme multi-label classification problem: a single patient note may contain multiple relevant codes among tens of thousands of possible labels. The distribution of ICD codes is typically long-tailed, where common chronic conditions such as hypertension or diabetes occur frequently, while countless rare conditions appear only a handful of times in training data (Rios and Kavuluru, 2018). In the MIMIC-III dataset, over 50% of codes never appear or occur fewer than ten times (Johnson *et al.*, 2016), making it challenging for models to learn robust representations for infrequent diagnoses.

This imbalance leads to a disparity in model performance: systems that do well on common labels often fail to recall rare codes. Edin et al. (Edin *et al.*, 2023) conducted a replicability study that exposed how certain train-test splits and suboptimal macro F1 calculations led to inflated performances for rare-code prediction. Their work emphasized fairer evaluation and transparent reporting of model limitations. Similarly, Nguyen et al. (Nguyen *et al.*, 2023) showed that MIMIC-IV features a more extreme code distribution, since it merges ICD-9 and ICD-10 subsets. Handling this imbalance remains a top priority, as rare codes can indicate serious or complex conditions requiring accurate documentation and billing.

Curriculum learning strategies (Ren *et al.*, 2022) partially address this issue by exploiting code hierarchies to share knowledge between parent and child labels. Additional techniques, such as oversampling and cost-sensitive learning, have also been tested, but they have been less effective at scale (Scheurwegs *et al.*, 2017). Consequently, novel approaches—particularly synthetic data augmentation—are garnering growing interest in mitigating the challenge of rare code prediction.

2.6.2 Processing Long and Complex Clinical Documents

Clinical narratives can be lengthy and may contain redundancies, inconsistencies, and a wide variety of clinical terminology. Early studies suggested that longer documents degrade model performance, but Edin et al. (Edin *et al.*, 2023) found the effect of length to be secondary to other factors such as code frequency. Nonetheless, managing long clinical notes remains computationally demanding.

Transformer-based models often face input length constraints (e.g., 512 or 1024 tokens), necessitating strategies such as truncation (Gao *et al.*, 2021), chunking (Huang *et al.*, 2022), or hierarchical encoding (Heo *et al.*, 2022). Although advanced architectures like Longformer or BigBird can theoretically handle up to 8k tokens or more, they are memory-intensive and can be slow to train. Document summarization and section-

based modeling present additional strategies but risk losing crucial detail about specific diagnoses. Finding scalable, efficient ways to represent entire discharge summaries remains an important research focus for real-world deployments.

2.6.3 Explainability and Interpretability

Explainability is essential in clinical applications, where clinicians, auditors, and coding professionals must trust automated coding outputs (Holzinger *et al.*, 2017). Attention-based methods, such as CAML (Mullenbach *et al.*, 2018) or LAAT (Vu *et al.*, 2020*a*), highlight relevant text segments for each predicted code, providing intuitive evidence for model decisions. However, these attention weights only offer correlational explanations—further research is needed to establish more causal or structured forms of explainability.

Edin et al. (Edin *et al.*, 2024) proposed an unsupervised approach that can achieve supervised-level explanation quality, suggesting that external knowledge or adversarial training may produce rationale annotations without costly manual labeling. Additionally, researchers have explored knowledge graphs and symbolic reasoning to justify individual code predictions (Teng *et al.*, 2020). Although these methods can yield greater transparency, fully interpretable models that seamlessly integrate domain knowledge remain an open challenge.

2.7 EMBEDDING MODELS AND CLINICAL SEMANTIC SEARCH

Embedding models underlie many of the text representation and retrieval tasks in healthcare. They transform sparse textual data into dense vectors that encode semantic relationships among words, sentences, or entire documents. Excoffier et al. (Excoffier *et al.*, 2024) compared generalist transformers (trained on broad corpora) to specialized clinical embeddings for ICD semantic search, observing that generalist embeddings were often more robust to minor text rephrasings. This finding implies that domain-specific pretraining might not always be superior if it narrows the model's domain to patterns

less frequent or less stylistically varied than the general language.

Nevertheless, domain-adapted models like ClinicalBERT, BioBERT, and PubMedBERT have achieved better performance in several medical NLP tasks, including ICD coding (Gao *et al.*, 2021; Ji *et al.*, 2021). The discrepancy in outcomes suggests that training set size, the nature of clinical text, and the distribution of code descriptions can all influence whether generalist or specialized embeddings hold the advantage. Moreover, specialized strategies such as code-synonym expansion (Yuan *et al.*, 2022) and concept-level attention highlight how embedding-based approaches can leverage existing biomedical ontologies (e.g., UMLS (Bodenreider, 2004), SNOMED CT (Donnelly, 2006), or cTAKES (Savova *et al.*, 2010)) to improve both performance and explainability.

2.8 KNOWLEDGE-BASED METHODS AND SYMBOLIC INTEGRATION

Knowledge-based methods aim to incorporate medical ontologies, taxonomies, and domain rules into deep learning architectures. These techniques can address shortcomings in purely data-driven models by providing explicit relationships between diseases or leveraging synonyms and hierarchical groupings (Xie *et al.*, 2019; Teng *et al.*, 2020). Such integration is especially valuable in ICD coding, where label space is inherently hierarchical, and textual evidence for certain diagnoses may be sparse.

Ren et al. (Ren *et al.*, 2022) demonstrated that hyperbolic embeddings capture hierarchical structure better than Euclidean embeddings, particularly for ICD codes. By representing codes in a hyperbolic space, parent-child relationships are learned with low distortion, and training proceeds from broader categories to finer levels. Similarly, Cao et al. (Cao *et al.*, 2020) proposed a hyperbolic co-graph representation that leveraged hierarchical information to enhance predictive performance. Knowledge-based approaches thus help unify symbolic reasoning with data-hungry neural models, thereby improving interpretability and filling gaps where training examples are limited.

2.9 SYNTHETIC DATA GENERATION FOR RARE ICD-10-CM CODES

2.9.1 Motivation and Techniques

The extreme class imbalance of ICD codes—where many rare diagnoses have few or no training examples—has motivated synthetic data generation as a complementary strategy to curriculum learning and data augmentation. Large Language Models (LLMs), such as GPT-3.5, GPT-4, or fine-tuned LLaMA, are increasingly employed to create realistic clinical narratives for underrepresented diagnoses (Kumichev *et al.*, 2024; Falis *et al.*, 2024). These synthetic notes preserve confidentiality while expanding the variety of text samples available for training models.

Falis et al. (Falis *et al.*, 2024) examined the feasibility of prompting GPT-3.5 with specific ICD-10-CM codes and descriptions to generate discharge summaries. Adding these synthetic notes to real training data produced modest performance improvements on rare codes and helped the model avoid out-of-family misclassifications. However, these gains came at the cost of potentially overfitting to the style of generated text, which might be less diverse than real clinical documentation.

Other augmentation methods include synonym replacement, back-translation, or adversarial generation of latent feature vectors, rather than raw text. Although these approaches are less computationally demanding than prompting large LLMs, they often provide smaller increases in coverage for rare labels. Hybrid strategies that combine knowledge-graph constraints with generative models have also been explored, ensuring that synthetic samples remain clinically plausible (Falis *et al.*, 2022).

2.9.2 Ethical and Practical Considerations

Synthetic data introduce potential advantages for privacy protection, as real patient identifiers can be omitted or obfuscated. Nevertheless, generative models might inadvertently reproduce sensitive or identifiable fragments if they encountered them during training. Ensuring high-fidelity clinical plausibility is also essential; simplistic or

repetitive patterns within synthetic samples could bias the model toward unrepresentative language (Pollard *et al.*, 2018).

In practice, synthetic notes appear to be most beneficial in few-shot or one-shot contexts, where the model has a handful of real examples for a rare code and gains additional variety through generation (Rios and Kavuluru, 2018). Full replacement of real data is typically detrimental. As a result, synthetic data remain a supplement rather than a primary information source, especially for regulating or auditing high-stakes billing environments.

2.10 BENCHMARKS AND EVALUATION METRICS

2.10.1 Public Datasets and Benchmarks

The research community has benefited significantly from openly available clinical datasets such as MIMIC-III (Johnson *et al.*, 2016) and MIMIC-IV (Johnson *et al.*, 2023). MIMIC-III focuses on ICD-9-CM codes, while MIMIC-IV provides a larger number of hospital admissions and reflects the transition to ICD-10-CM coding (Nguyen *et al.*, 2023). Both datasets contain lengthy discharge summaries with manually assigned ICD codes, making them ideal for evaluating multi-label classification performance. The eICU Collaborative Research Database (Pollard *et al.*, 2018) supplies additional notes and structured data across multiple hospital systems, enabling analyses of model generalizability.

Benchmark initiatives standardize tasks, train-dev-test splits, and evaluation pipelines. This fosters comparability among different architectures (e.g., CNNs vs. BERT vs. hierarchical models) and clarifies progress in rare-code prediction. Replication studies have demonstrated that small variations in preprocessing or code filtering can materially affect reported metrics, emphasizing the need for consistent benchmark protocols (Edin *et al.*, 2023).

2.10.2 Common Evaluation Metrics

Because automated ICD coding is a multi-label task, precision and recall must be balanced to capture different aspects of performance:

- **Micro-F1**: Aggregates all predicted and true labels across the dataset, giving higher weight to frequently occurring codes. This is typically higher than macro-F1 and often used to illustrate a model's overall performance on the bulk of common codes.

- **Macro-F1**: Computes the F1-score per code and then averages across all codes, giving equal importance to rare and common labels. Macro-F1 is sensitive to performance on the long tail of infrequent codes and is usually much lower.

- **Micro/Macro AUC**: Measures the area under the ROC curve, either pooling predictions across codes (micro) or taking a simple average of per-code AUC (macro). High AUC often indicates that, with careful threshold tuning, the model could achieve good recall and precision.

- **Precision@K**: Assesses how many of the top K predicted codes match the ground truth. This metric is sometimes used in practical coding assistance tools to limit the suggestions displayed to human coders.

- **Hierarchical Metrics**: Account for partial correctness if a model predicts a closely related code from the same family, as opposed to a completely unrelated code. Such metrics better reflect real-world coding tasks, where distinguishing the correct subtype can be secondary to identifying the right disease category.

Error analyses often examine whether incorrect predictions are at least within the correct organ system or parent node in the ICD hierarchy. Some studies also evaluate interpretability by checking if highlight-based explanations align with key phrases annotated by clinical experts (Mullenbach *et al.*, 2018). These deeper analyses provide a more nuanced picture of model behavior than a single F1 score.

2.11 FUTURE DIRECTIONS AND OPEN PROBLEMS

Despite the substantial progress described thus far, automated ICD coding continues to face many open challenges:

Self-Supervised and Domain-Adaptive Pretraining. The majority of current models rely on supervised training with limited labeled data. Future systems may exploit large unlabeled clinical corpora through advanced self-supervised objectives or domain-adaptive methods (Gururangan *et al.*, 2020). Domain-specific LLMs could capture complex medical language better than general-purpose ones, potentially reducing reliance on manually annotated data.

Few-Shot and Zero-Shot ICD Coding. The long-tailed distribution of ICD codes motivates the development of architectures that can accurately code new or rare labels with minimal training examples (Rios and Kavuluru, 2018; Yang *et al.*, 2023). Models that incorporate code definitions or hierarchical relations—along with advanced prompting techniques—could infer rare labels even when they appear infrequently or not at all in training data.

Multimodal Integration. Clinical decision-making involves not only unstructured text but also structured data (e.g., labs, vitals) and imaging findings (Choi *et al.*, 2016). Effective multimodal architectures that combine textual discharge summaries with time-series or image-based signals remain under-explored but hold potential for more holistic coding.

Interpretability and Interactive Learning. Although attention-based and highlight-driven methods provide partial transparency, future research may explore causal explanations, prototype-based retrieval, or rationale generation to reinforce clinicians' trust. Interactive AI coding assistants could prompt human coders for clarification, use that feedback to refine predictions, and thus achieve greater accuracy over time.

Ethical and Regulatory Considerations. Clinical data are sensitive, and large-scale LLM usage must comply with data protection laws. Synthetic generation can mitigate privacy concerns but raises questions about realism and potential bias. Regulatory acceptance may require robust auditing mechanisms, reproducible evaluation, and evidence that automated coding does not compromise patient safety or billing integrity.

ICD-11 and Beyond. The introduction of ICD-11 presents new structural and semantic complexities. Transferring knowledge from ICD-9 or ICD-10 to ICD-11 demands flexible architectures capable of adapting to new taxonomies. Approaches that use hierarchical embeddings and code-description modeling can help manage these shifts in labeling systems (Gaebel *et al.*, 2020).

2.12 SUMMARY OF THE LITERATURE

Automated ICD coding presents a set of distinct technical and organizational challenges. Initial studies employed rule-based logic and feature-engineered machine learning but were constrained by scalability and the evolving nature of clinical vocabularies (Scheurwegs *et al.*, 2017). Deep neural architectures, particularly CNNs, RNNs, and transformers, provided significant accuracy improvements by learning hierarchical text representations without extensive manual engineering (Mullenbach *et al.*, 2018; Li and Yu, 2020; Heo *et al.*, 2022).

Transformer-based models and newer BERT-like encoders excel at capturing complex linguistic features in discharge summaries. However, they require careful handling of long documents, face persistent issues with rare diagnoses, and demand large computational resources. Curriculum learning, hierarchical embeddings, and synthetic data generation offer partial remedies, enhancing performance on uncommon codes and improving interpretability through structured knowledge infusion (Ren *et al.*, 2022; Falis *et al.*, 2024).

Recent benchmarking efforts clarify that future progress lies in robust domain-specific pretraining, advanced handling of extreme multi-label imbalances, and deeper integration of medical ontologies. Additionally, evolving demands—such as ICD-11 coding—underline the importance of flexible systems that adapt to changing taxonomies. The development of interactive, interpretable coding assistants, combined with rigorous evaluation and continuous learning, is expected to guide the field toward reliable real-world implementation in clinical and administrative workflows.

CHAPTER 3

PROBLEM DEFINITION AND FORMULATION

3.1 SCOPE OF THE PROBLEM

Accurate labeling of clinical notes with *International Classification of Diseases (ICD)* codes is recognized as a critical task in contemporary healthcare documentation. Clinical narratives, however, often manifest as lengthy, highly unstructured text that contains abbreviations, domain-specific terminology, and nuanced clinical indicators. A further complication arises when only a small fraction of ICD codes—here termed "rare codes"—occurs so infrequently that very few labeled instances are available for machine learning models. These codes frequently remain under-detected, which can diminish overall coding accuracy and potentially undermine large-scale health analytics. The objective of the present work is to automate the assignment of *multiple ICD codes* to each clinical document, with a specific focus on *improving predictive effectiveness for rare labels*.

Failure to correctly classify clinical narratives or to assign all relevant labels can have far-reaching consequences. At the institutional level, coding errors may result in imprecise billing processes and resource misallocation. At the individual level, a missed or erroneous diagnosis code may obscure a patient's condition or hamper effective disease management. On a broader scale, large-scale analytics that rely on aggregated administrative data may not capture the true incidence or prevalence of rare medical conditions, causing gaps in research and healthcare policy.

3.2 CENTRAL CHALLENGES

Although recent advances in machine learning have significantly improved automated ICD coding, several formidable obstacles persist:

- **Long-Tail Distribution:** The majority of available data is dominated by a few frequent codes, leaving rare diagnoses severely underrepresented.

- **High-Dimensional Label Space:** Each clinical document may include multiple ICD codes, and semantic interdependencies among these labels complicate multi-label classification.

- **Textual Complexity:** The unstructured nature of clinical notes, featuring inconsistent formatting and heavy use of abbreviations and domain-specific jargon, complicates text processing.

- **Knowledge Integration:** Existing ontologies contain valuable domain knowledge, but straightforward classification pipelines often overlook hierarchical or semantic relationships among medical codes.

- **Explainability Needs:** Models designed for clinical applications must be comprehensible to end-users, who require transparency and trust when evaluating automatic code assignments.

These challenges pose general difficulties for all ICD codes, but the detrimental effects are exacerbated for rare labels. Systems that perform well for frequently observed diagnoses often fail to capture low-frequency conditions, an omission that motivates the development of strategies tailored to address class imbalance and explainability gaps.

3.3 MATHEMATICAL FORMULATION OF THE TASK

A concise mathematical formulation is presented below to clarify how a trained model should transform unstructured clinical text into accurate ICD predictions.

3.3.1 Defining the Space of Clinical Narratives

Let $\mathcal{X}$ represent the collection of all possible clinical narratives. Each element $x \in \mathcal{X}$ is a textual record describing a patient's clinical episode. In practice, x may include:

- *Admission notes* reflecting the circumstances motivating a patient's hospitalization.

- *Discharge summaries* documenting final diagnoses, interventions, and follow-up steps.

- *Progress notes* covering daily clinical updates for extended hospital stays.

Despite a variety of internal structures and formats, each x can be treated as a string of

tokens (or words) for classification purposes. Thus, the set $\mathcal{X}$ defines the *input space* over which the model operates.

3.3.2 Defining the Label Space of ICD Codes

Let $\mathcal{L} = \{\ell_1, \ell_2, \ldots, \ell_{|\mathcal{L}|}\}$ be the full set of ICD codes. The notation $|\mathcal{L}|$ denotes the number of distinct codes, which can be large in real-world applications:

- Major ICD revisions (ICD-9, ICD-10, ICD-11) contain thousands of codes.
- Many codes represent specialized conditions (subtypes or sub-phenotypes).

Since multiple diagnoses are commonly documented for a single admission, the label space is multi-label in nature, allowing each clinical document to be assigned any subset of $\mathcal{L}$.

3.3.3 Labeled Dataset Construction

Assume that a set of N labeled clinical narratives is available:

$$D = \{(x_i, Y_i)\}_{i=1}^{N},$$

where $x_i \in \mathcal{X}$ and $Y_i \subseteq \mathcal{L}$. Each pair (x_i, Y_i) consists of a discharge summary (or similar record) and the corresponding ICD codes assigned by professional coders. This dataset D forms the basis for supervised learning.

3.3.4 Function Definition

The goal is to learn a function

$$f : \mathcal{X} \longrightarrow 2^{\mathcal{L}}, \tag{3.1}$$

where $f(x) = \hat{Y}$ for $\hat{Y} \subseteq \mathcal{L}$. Here, $2^{\mathcal{L}}$ represents the power set of $\mathcal{L}$. For any clinical document x, the prediction $\hat{Y}$ is the subset of codes that the model deems relevant. In principle, $\hat{Y}$ can be empty, although that outcome is uncommon in practical medical scenarios.

In most current architectures, a multi-label classification approach is adopted. Specifically,

the model generates a probability or score for each code in $\mathcal{L}$, and thresholding (or a related method) is then applied to map those scores to a predicted subset.

3.3.5 Objective: Composite Metric Maximization

A composite performance metric Ω is introduced to reflect the importance of both overall correctness and the accurate identification of rare labels. In formal terms, the aim is:

$$\underset{f}{\text{maximize}} \quad \Omega(\text{Accuracy of } f,\ \text{Rare-Codes Metrics}). \tag{3.2}$$

Depending on institutional priorities, Ω may combine measures such as micro-F1 or micro-AUC (evaluating overall performance across all codes) with macro-F1 or a specialized metric that focuses on rare codes. A weighted sum or a multi-objective formulation can be employed. This dual emphasis on both frequent and rare diagnoses ensures that gains do not come at the expense of clinically significant but infrequent conditions.

3.4 SIGNIFICANCE OF RARE CODES

Rare codes pose a particular challenge due to limited representation in real-world data, yet these codes often signal clinically important conditions.

3.4.1 Definition and Challenges

A rare code may appear only a few times in a large dataset. Let $\mathcal{L}_{\text{rare}} \subseteq \mathcal{L}$ be the set of codes whose frequencies fall below a chosen threshold τ:

$$\mathcal{L}_{\text{rare}} = \{\ell \in \mathcal{L} \mid \text{Frequency}(\ell) < \tau\}.$$

Typical values for τ might be a fixed count (e.g., 50) or a percentile (e.g., the bottom 10%). Learning to detect rare codes is complicated by these labels' scarcity in training data and the specialized language often observed in notes that reference such conditions.

Overall loss functions tend to favor frequent labels, marginalizing rare ones.

3.4.2 Impact of Rare Code Misclassification

Misclassification of rare diagnoses can produce substantial adverse effects:

- **Clinical Impact:** A rare but critical diagnosis, if missed by an automated system, may affect patient care when clinicians rely on coded data for decision support.
- **Research Bias:** Underrepresentation of rare conditions in large-scale analyses can mask the true disease burden.
- **Billing and Compliance:** Inaccurate coding for rare conditions can lead to underpayment or claim denial, as well as potential auditing complexities.

Consequently, rare code detection forms a key component of robust automated coding systems.

3.4.3 Recall and Precision Constraints for Rare Codes

For a clinical note x with true codes Y, where $Y \cap \mathcal{L}_{\text{rare}} \neq \emptyset$, recall and precision for the rare portion are of particular concern. Low recall implies a failure to capture rare conditions, whereas low precision may yield an excess of false positives for these labels. Either error type can be problematic in a medical context, emphasizing the importance of balanced metrics.

3.4.4 Analytical Example of Class Imbalance

A simple numerical illustration can clarify the issue. Consider 5,000 labeled notes, and suppose:

- A frequent code ℓ_{common} appears in 1,200 notes.
- A rare code ℓ_{rare} appears in 20 notes.

A classifier that never predicts ℓ_{rare} incurs only 20 false negatives on a base of 5,000, so the impact on the global accuracy may be less than 1%. Such a result can inadvertently encourage ignoring rare codes. This outcome underscores the need for targeted strategies addressing low-frequency classes.

3.5 PROPOSED METHODOLOGICAL FOCUS

Several methodological approaches are planned to mitigate performance deficits in both frequent and rare ICD codes:

1. **Data Augmentation for Rare Codes:** Synthetic data generation or specialized oversampling strategies that boost representation of low-frequency labels.

2. **Advanced Multi-Label Architectures:** Network designs that incorporate hierarchical embeddings and structured label representations, taking advantage of semantic relationships within the ICD taxonomy.

3. **Integration of External Knowledge:** Leveraging medical ontologies (e.g., SNOMED, UMLS) to enhance label embeddings and encourage clinically consistent classification.

4. **Model Interpretability:** Developing explainable frameworks where predicted ICD codes can be traced back to their textual justifications, thereby fostering user trust.

Each of these elements seeks to address the data imbalance inherent in ICD coding tasks while preserving or improving performance on all relevant labels, including those with scarce training instances.

3.6 EVALUATION FRAMEWORK

Evaluation in this domain requires a range of measures to capture overall performance and rare-code outcomes:

- **Micro-F1**: Reflects globally aggregated precision and recall.

- **Macro-F1**: Averages precision and recall across codes, thereby placing additional emphasis on rare labels.

- **Recall@k**: Indicates how effectively the model recovers correct codes in its top-k predictions, a practical measure for coding assistance tools.

- **Rare-Code Metrics**: Tracks performance (e.g., precision, recall, F-score) on the subset $\mathcal{L}_{\text{rare}}$, ensuring improvements are realized for both frequent and rare labels.

A balanced presentation of both micro- and macro-level metrics is essential, given the skewed distribution of ICD codes. It is possible for a model to exhibit strong micro-F1

yet fail to detect many rare diagnoses, which would be revealed by a lower macro-F1 or poor rare-code metrics.

3.7 CONSTRAINTS AND OBJECTIVES

Practical constraints and overarching objectives guide this research:

- **Data Privacy**: Clinical text is sensitive, and synthetic data or augmentation techniques must preserve confidentiality.
- **Computational Resources**: Many clinical institutions have limited hardware, thus requiring solutions that balance accuracy with feasible computational costs.
- **Transparency and Adoption**: Models must present interpretable outputs so that clinical personnel can audit and trust the system, particularly for high-stakes or unfamiliar diagnoses.

Consequently, the proposed methods will be designed to accommodate these practical considerations, alongside the main objective of enhancing performance for both common and infrequent ICD codes.

CHAPTER 4

METHODOLOGY

4.1 INTRODUCTION AND SCOPE

This chapter sets forth a *multi-phase methodology* aimed at refining automated ICD-10-CM code prediction, *especially* for rare codes. A comprehensive framework is proposed, featuring an *ontology-based disease grouping* strategy, a pipeline for *synthetic discharge summary generation*, a *hybrid model training* approach that integrates both real and synthetic data, and an evaluation scheme specifically *attuned* to rare code distribution imbalances. A *modular and extensible workflow* is emphasized, so that the methodology may be adapted readily to alternative scenarios involving imbalanced classification. Ultimately, validation is planned on a real-world clinical dataset to assess the *generalizability* and *utility* of this framework.

4.2 ONTOLOGY-BASED DISEASE GROUPING

Rare ICD-10-CM codes frequently exhibit *extremely low prevalence*, which hinders the ability of predictive models to learn discriminative features. In order to mitigate this challenge, an *ontology-based grouping* approach has been envisioned. This process entails parsing established medical ontologies—such as *SNOMED CT*, *UMLS*, and the *Orphanet Rare Disease Ontology*—to merge *hierarchical* and *associative* relationships.

It is intended that each rare disease will be anchored in a *knowledge graph*, and *closely related* concepts will be identified via direct graph traversal. These items will be clustered into a *unified disease family*, thereby providing a *contextual* lens through which the rare disease can be understood. By associating each rare disease with more common conditions, the model will receive exposure to realistic clinical context, improving its capacity to recognize the *linguistic cues* relevant to rare diagnoses. Additionally, ontology

mapping ensures that no *orphaned codes* are disregarded, given that each disease is matched at least to a more general ICD-10-CM code if an exact match proves elusive.

It is anticipated that this ontology-centric approach will enable more *robust learning* for rare codes, because they will be presented alongside clinically related diagnoses. By placing rare codes in *medically coherent families*, the methodology reduces the isolation that often occurs when codes appear only a handful of times in conventional datasets.

4.3 SYNTHETIC DISCHARGE SUMMARY GENERATION

Because rare codes remain *infrequent* in standard patient records, a *synthetic data generation* process is introduced to amplify the presence of these *underrepresented labels*. A retrieval-augmented generation pipeline has been planned, wherein medical facts, professional guidelines, and relevant textual fragments are *retrieved* to guide a large language model in producing draft discharge summaries.

- **Structured Prompting:** Placeholders will be embedded for sections such as *Chief Complaint*, *History of Present Illness*, *Hospital Course*, *Medications*, and *Discharge Diagnoses*.
- **Disease Clusters:** Rare diseases will be linked with related diagnoses identified through ontology-based grouping, ensuring that *realistic co-occurrences* appear in the synthetic text.
- **Multi-Agent Feedback Loop:**
  - A *draft discharge summary* is generated by the language model.
  - A *critic agent* then examines the draft for logical inconsistencies and *missing details*.
  - Iterative refinements are made until the note meets *plausibility* and *consistency* criteria.

The final *synthetic discharge summaries* must provide sufficient evidence for the relevant rare ICD-10-CM codes, mirroring the style and structure of authentic clinical documentation. This balanced approach is expected to *significantly* enrich the training

dataset for rare codes, thereby reducing distributional skew and enhancing overall model recall for these infrequent conditions.

4.4 DEEP LEARNING MODEL TRAINING AND TESTING

An *augmented training corpus* composed of both real and synthetic discharge summaries will subsequently be employed in training two deep learning architectures: (i) a *transformer-based* model, and (ii) a *CNN-based* model. The transformer-based model is *anticipated* to capture *long-range dependencies*, whereas the CNN-based model should excel at identifying *localized textual patterns* pertinent to specific codes.

A multi-label classification scheme will be used, assigning probabilities to all potential ICD-10-CM codes. To address *class imbalance*, strategies such as *weighted sampling*, *focal loss*, and *class-weighted loss* have been planned. Synthetic summaries will be *oversampled* or *weighted* more heavily, ensuring that rare codes appear frequently enough in training batches. The training procedure will commence with *binary cross-entropy* and moderate class weighting, followed by the incremental introduction of *focal loss* during later epochs, thus allowing the model to *fine-tune* its attention on the *most difficult* labels.

A development set containing both real and synthetic notes will guide *hyperparameter tuning* and *early stopping*. To confirm the *practical value* of this method, the final evaluation will be conducted on a *held-out portion* of real MIMIC-IV discharge summaries, assessing whether improvements in *rare code classification* carry over to genuine patient data.

4.5 EVALUATION METRICS AND ANALYSIS

The planned evaluation seeks to capture *various dimensions* of performance, with special focus on rare code detection. Several metrics will be reported, each *illuminating* a

different aspect of classifier efficacy. The following formulas present the principal metrics in detail (where TP = true positives, FP = false positives, FN = false negatives, and C = total number of classes):

1. **Micro-F1**

$$\text{Micro-F1} = 2 \times \frac{\sum_{i=1}^{C} TP_i}{\sum_{i=1}^{C} (2 \cdot TP_i + FP_i + FN_i)} \tag{4.1}$$

   Micro-F1 aggregates the contributions of all codes to compute the overall precision and recall across the entire label space.

2. **Macro-F1**

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^{C} \text{F1}_i \tag{4.2}$$

   Each class-level F1 score F1_i is calculated, and their average is taken to highlight how the model performs on *each individual* label, giving rare codes equal weight to common codes.

3. **Recall@*k***

$$\text{Recall@k} = \frac{\text{Number of times the correct code is in the top } k \text{ predictions}}{\text{Total number of samples}} \tag{4.3}$$

   This metric indicates the fraction of cases in which the *true* codes appear among the top k predictions, a setting relevant in *coding-assistance systems* that present a limited list of suggestions.

4. **Exact Match Ratio**

$$\text{Exact Match} = \frac{\text{Number of notes for which the full set of codes is correctly predicted}}{\text{Total number of notes}} \tag{4.4}$$

   This stricter criterion gauges the extent to which all predicted codes align *perfectly* with the ground truth set for each note.

5. **Never-Predicted Rate**

$$\text{Never-Predicted Rate} = \frac{\text{Number of codes that are never predicted by the model}}{C} \tag{4.5}$$

   This measure highlights whether any codes—particularly *rare ones*—are completely ignored.

Emphasis will be placed on determining whether the *ontology-based grouping* and *synthetic data* augmentations produce measurable gains in macro-F1 and reduce the *never-predicted rate*. A targeted error analysis will be conducted for codes that remain

unrecalled, providing directions for further refinements.

4.6 PROCESSING MIMIC-IV AS OUR MAIN TEST DATASET

In this work, the MIMIC-IV database will be employed as the principal test dataset for automated ICD coding. This dataset contains de-identified clinical information collected from the Beth Israel Deaconess Medical Center (BIDMC), with particular emphasis on structured hospital records and unstructured discharge summaries. The following steps describe how the data will be prepared, managed, and validated in a manner consistent with privacy regulations and best practices in clinical data science.

4.6.1 Data Extraction and Integration

The relevant tables from MIMIC-IV (e.g., `admissions`, `patients`, `diagnoses_icd`, and `discharge`) will be identified and retrieved. These tables will then be joined on keys such as `subject_id` and `hadm_id`, thus creating a unified view of patient admissions, annotated diagnoses, and the corresponding discharge narratives. During this process, all admissions lacking a corresponding discharge summary or valid ICD-10 code assignments will be flagged for potential exclusion, ensuring that the resulting dataset remains suitable for supervised learning.

4.6.2 Text Standardization and Cleaning

Each discharge summary will be examined to remove any remaining placeholders or noise introduced by the de-identification process. Lowercasing and tokenization will be conducted if deemed beneficial for subsequent analyses, and the presence of medical abbreviations or numeric identifiers will be noted. Segmentation into sections (such as “History of Present Illness” and “Medications”) may be carried out to reduce text complexity, though the entire narrative will remain intact if full-length modeling is desired.

4.6.3 Handling Duplicates and Missing Data

Instances in which multiple, nearly identical discharge summaries exist for a single admission will be managed by detecting duplicates through textual similarity checks. These checks will reduce the risk of overrepresenting particular patients or hospital stays. Any admissions without sufficiently long or clearly documented notes will be handled through either imputation strategies or careful removal, thereby minimizing noise in model training and evaluation.

4.6.4 Code Distribution and Rare Labels

A preliminary examination of the ICD-10 codes assigned to each admission will be performed to understand their frequency distribution. Since ICD codes often follow a long-tail pattern, additional attention will be devoted to codes that appear infrequently. If these rare labels contribute significantly to the overall error rate, strategies such as targeted oversampling or retrieval-augmented data enrichment may be considered, provided such techniques respect the constraints of de-identification and data plausibility.

4.6.5 Database Management and Version Control

All data transformations and table joins will be executed within a controlled environment (e.g., DuckDB) to allow for efficient querying and seamless integration with Python-based analytic workflows. The resulting datasets, along with any derived tables, will be tracked using version control to ensure that incremental changes are systematically logged and reproducible. This practice will enable transparent experimentation and simplify the process of rolling back to earlier dataset states if necessary.

4.6.6 Security and Compliance

All patient information in MIMIC-IV is protected via de-identification, and no direct identifiers (e.g., names, addresses) are present. It will be confirmed that the data use agreement from PhysioNet is upheld, particularly regarding re-identification risks and data sharing restrictions. Any synthetic expansions or textual outputs generated by the

pipeline will be carefully reviewed to ensure that confidential details are not reintroduced.

CHAPTER 5

DATASET UNDERSTANDING

5.1 OVERVIEW OF MIMIC-IV

MIMIC-IV is a large, openly accessible database of de-identified electronic health records (EHR), developed through a collaboration between the Beth Israel Deaconess Medical Center (BIDMC) and the Massachusetts Institute of Technology (MIT). It spans clinical encounters from 2008 to 2019 and contains comprehensive data, including demographic information, laboratory values, prescriptions, billing codes, clinical notes, and more. The dataset is organized into multiple modules to reflect different sources or systems:

- `hosp`: Hospital-wide data, containing tables such as `admissions`, `patients`, and `diagnoses_icd`.
- `icu`: Intensive Care Unit (ICU) data with highly granular time-series measurements (e.g., `chartevents`).
- `ed`: Emergency department records.
- `note`: De-identified clinical notes, such as discharge summaries.
- `cxr`: Information linking chest X-ray images to the relevant patients in MIMIC-CXR.

For the purpose of analyzing discharge summaries and automatically assigning ICD codes, particular emphasis is placed on:

- `patients` in the *hosp* module, which provides demographics such as `anchor_age` and `anchor_year_group`.
- `admissions` in the *hosp* module, which contains `hadm_id` (hospital admission ID) and timestamps (`admittime`, `dischtime`).
- `diagnoses_icd` in the *hosp* module, indicating both ICD-9 and ICD-10 billed diagnoses.
- `discharge` in the *note* module, offering the textual discharge summaries that are

crucial for modeling.

All queries used to gather counts, distributions, and other information about the MIMIC-IV dataset are provided in the Appendix (Chapter A). In the following sections, key outcomes from these queries are summarized, and visual explorations are presented.

5.2 SQL QUERY RESULTS AND DATA HIGHLIGHTS

5.2.1 Patient and Admission Volumes

Using the queries referenced in Figures A.1 and A.2 in the Appendix, it was determined that there are **364,627 unique patients** in the `patients` table. In the `admissions` table, there are **546,028 unique admissions**.

5.2.2 Multiple Admissions per Patient

Figure A.3 in the Appendix provides the SQL code used to examine how many admissions each patient has. Some patients experience over 100 admissions, highlighting that MIMIC-IV can capture highly recurrent hospital visits for certain individuals.

5.2.3 Discharge Summaries

By referencing Figure A.4 in the Appendix, it was found that there are **331,793** discharge summaries (`discharge` table). Nearly 214,296 admissions in MIMIC-IV do not have a final discharge note (see Figure A.5), underscoring that not every hospital stay yields a lengthy textual record.

5.2.4 Characterizing Note Length

The query in Figure A.6 computed word-count statistics for discharge summaries. The median note length is approximately 1,556 words, and at the 99th percentile, notes can exceed about 3,986 words. This wide variation in length indicates that some admissions have minimal documentation, whereas others include very detailed records.

5.2.5 ICD-9 vs. ICD-10 Utilization

Figure A.7 in the Appendix confirms that MIMIC-IV contains approximately 3.46 million ICD-10 code assignments and 2.91 million ICD-9 code assignments. This distribution suggests that modern coding practices (ICD-10) have become more prevalent. Detailed counts of top ICD-10 codes were retrieved using the query in Figure A.8, which revealed chronic conditions like hyperlipidemia (`E785`) and hypertension (`I10`) as highly common.

5.2.6 Long-Tailed Code Distribution

A query shown in Figure A.9 was used to confirm that there are **19,440 unique ICD-10 codes** in the dataset. As observed through the query in Figure A.10, around 9,217 ICD-10 codes appear fewer than five times, illustrating the classic long-tail distribution. Models must be robust to rare code occurrences.

5.2.7 Codes Per Admission

The query in Figure A.11 determined that there are on average $\sim$ 13.6 ICD-10 codes billed per admission. Consequently, this setting represents a multi-label classification task: each discharge summary can be associated with numerous codes rather than a single label.

5.3 VISUAL EXPLORATIONS OF MIMIC-IV

Plot 1: Distribution of Anchor Age

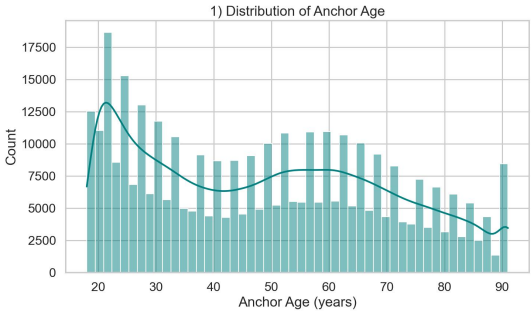

Figure 5.1: A histogram revealing a right-skewed distribution of anchor ages, with many individuals in their 20s and 30s.

Figure 5.1 shows how anchor ages are concentrated in younger to middle-aged adults, with a tail that extends to the maximum capped age of 91. There is a gradual decrease in frequency after middle age, but a mild uptick near 90–91, reflecting censoring of ages above 89.

Plot 2: Boxplot of Anchor Age by Admission Type

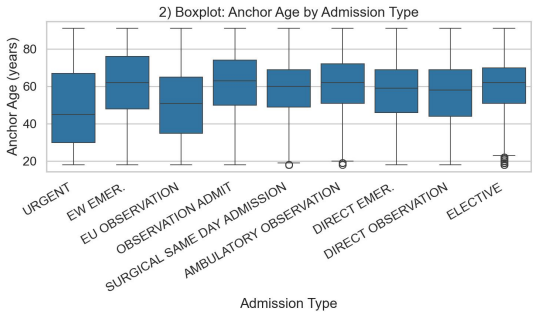

Figure 5.2: A boxplot illustrating how anchor age varies across different admission types.

Figure 5.2 offers insight into differences in age distributions for various hospital admission pathways (e.g., emergency admissions vs. elective surgeries). Some admission types span a broader age range, while others are more narrowly distributed, reflecting the patient populations associated with particular clinical services.

Plot 3: Admissions by Anchor Year Group

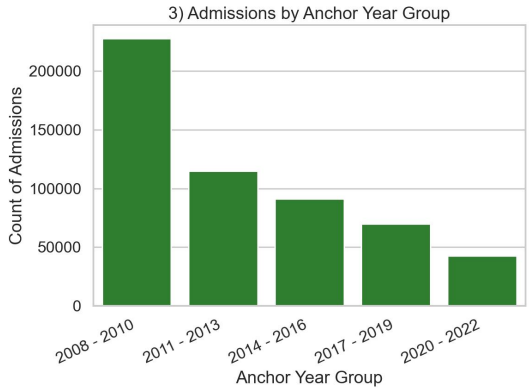

Figure 5.3: A bar chart showing the number of admissions categorized by approximate calendar year groups.

Figure 5.3 demonstrates a higher number of admissions in the earlier year intervals (2008–2010, 2011–2013) relative to later groups, likely reflecting the data collection periods and any changes in hospital practices over time.

Plot 4: Top 15 ICD-10 Codes by Frequency

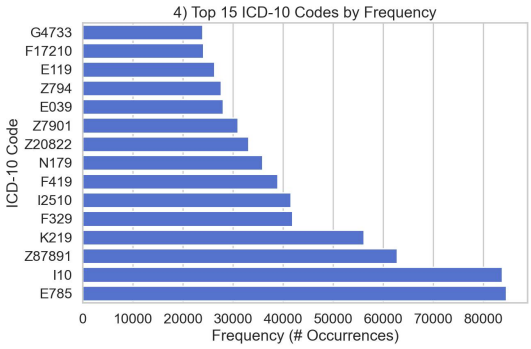

Figure 5.4: A horizontal bar chart showing the most common ICD-10 codes among admissions.

In Figure 5.4, chronic conditions such as hyperlipidemia (`E785`) and hypertension (`I10`) are clearly visible at the top of the frequency distribution, demonstrating the widespread prevalence of these diagnoses.

Plot 5: Co-Occurrence Matrix of Top 15 ICD-10 Codes

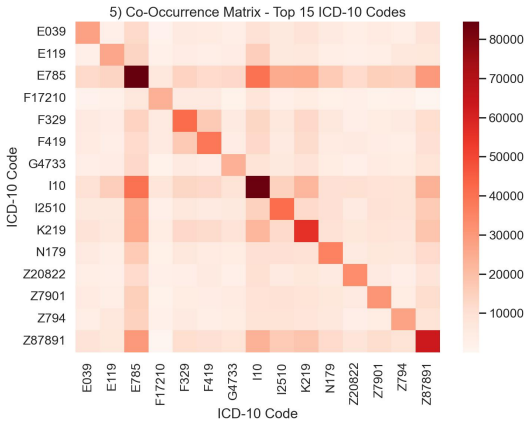

Figure 5.5: A heatmap illustrating how frequently pairs of the top 15 ICD-10 codes co-occur within the same admission.

Figure 5.5 reveals strong comorbidity patterns among high-frequency diagnoses. For instance, codes associated with hyperlipidemia often co-occur alongside hypertension, reflecting typical chronic disease clusters.

Plot 6: Discharge Note Length vs. Number of ICD-10 Codes

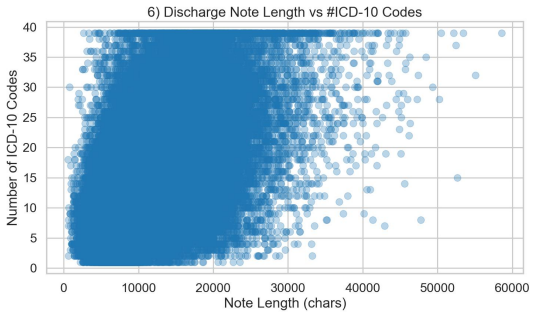

Figure 5.6: A scatter plot highlighting the relationship between note length (in characters) and the number of ICD-10 codes.

Figure 5.6 demonstrates a positive correlation: admissions with more diagnoses often generate longer discharge summaries. However, many cases are concentrated below 20,000 characters, indicating that moderately sized notes can still have multiple codes.

Plot 7: Violin Plot of Code Counts by Anchor Year Group

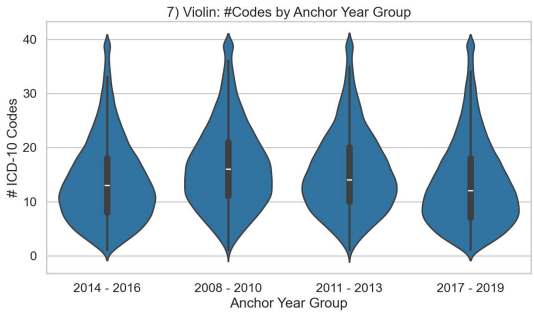

Figure 5.7: A violin plot representing how the distribution of total ICD-10 codes per admission varies across anchor year groups.

Figure 5.7 visualizes the spread of ICD-10 code counts for each time interval. Wide tails indicate that certain admissions receive a very large number of codes, reflecting complex clinical circumstances.

Plot 8: Mean Discharge Note Length by Anchor Year Group

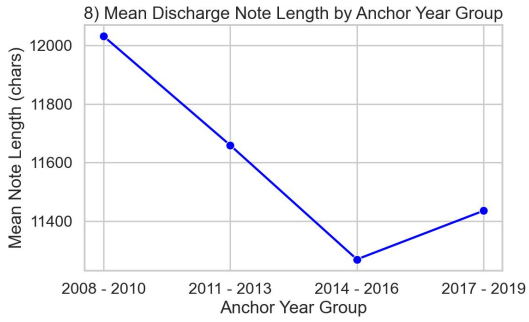

Figure 5.8: A line plot tracking average discharge summary lengths over five anchor year group intervals.

Figure 5.8 suggests that longer average notes were generated during the earlier collection periods (2008–2010), followed by a dip in 2014–2016 and a partial rebound in later intervals. Evolving clinical documentation practices may contribute to these shifts.

Plot 9: Top 10 Admission Types by Volume

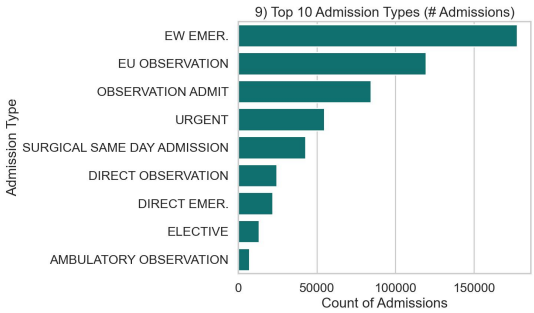

Figure 5.9: A horizontal bar chart showing the top ten admission types (e.g., emergency, elective, urgent).

Figure 5.9 highlights the prominence of emergency or unplanned admissions. Emergency admissions dominate, but there are also notable volumes in scheduled categories such as elective surgeries.

Plot 10: ICD-10 Code Frequency by Anchor Year Group (Top 15 Codes)

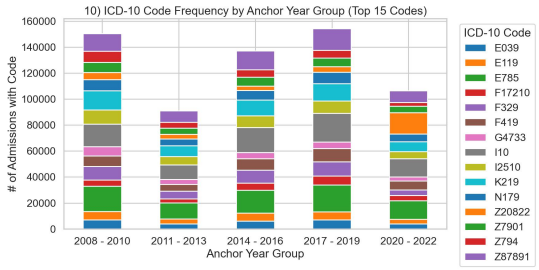

Figure 5.10: A stacked bar chart displaying the occurrence of each top-15 ICD-10 code within different anchor year groups.

Figure 5.10 shows that certain diagnoses remain consistently frequent across all anchor year intervals, again reflecting the prevalence of long-standing chronic conditions such as metabolic disorders and cardiovascular disease.

5.4 KEY INSIGHTS

From these queries and visualizations, several points emerge:

- **Large-Scale Data:** There are more than 364k unique patients and over 546k admissions, providing robust coverage of a hospital population.
- **Multiple Diagnoses:** The average admission carries more than 13 ICD-10 codes, confirming a multi-label setting.
- **Long-Tail Challenge:** Over 9,217 ICD-10 codes appear fewer than five times, highlighting difficulties with rare or specialized diagnoses.
- **Varied Note Lengths:** Discharge summaries range from brief texts to lengthy narratives with thousands of words.

5.5 FINAL DATA PREPARATION

For the goal of building a text-to-ICD coding model (akin to AWS Comprehend Medical or other automated medical coding applications), a consolidated table was created that merges each admission's discharge note with the set of associated ICD-10 codes. The

approach takes each row in the `discharge` table, then joins it with the corresponding entries in `diagnoses_icd` for that `hadm_id`.

The query used to accomplish this is shown in Figure A.12 in the Appendix. Each row in the resulting table contains:

- **notes**: The original discharge summary (raw text).
- **icd_codes**: A comma-separated list of all ICD-10 codes assigned to that admission.

This design directly supports multi-label classification, as the full narrative of the discharge summary is provided (with no additional demographic or structural metadata) alongside all billed ICD-10 codes. Such a format simulates real-world scenarios in which only free-text documentation is available for downstream coding or NLP tasks.

CHAPTER 6

RESULTS

We start with the outcomes of two preliminary investigations conducted to establish initial benchmarks and insights for automated ICD coding on the MIMIC-IV dataset. A simple *K-Nearest Neighbors (KNN)* model was first implemented to observe how a straightforward algorithm manages the complexity of clinical text classification. Subsequently, a *MinHash-based* similarity detection method was employed to reveal instances of near-duplicate discharge summaries and to estimate potential redundancy in the dataset. By examining these findings through a first-principles lens, the underlying causes—*data imbalance, text intricacy, and recurring clinical narratives*—emerge clearly, along with their effects on model performance and dataset quality.

6.1 BASELINE MODEL: K-NEAREST NEIGHBORS (KNN)

A KNN classifier was trained on discharge summaries that had been transformed into a TF-IDF vector space (limited to 5,000 features and incorporating both unigrams and bigrams). The `MultiLabelBinarizer` function was employed to accommodate the assignment of multiple ICD codes to each note. By design, KNN relies on calculating distances in the feature space to assign labels. However, due to the *intrinsic imbalance of ICD codes* and the *high-dimensional nature of clinical narratives*, the model's ability to generalize was substantially constrained. This constraint directly resulted in a *Macro F1-Score of 0.12*, revealing that frequent codes were mostly captured, whereas rare codes were systematically overlooked. The pronounced Hamming Loss further illustrated that the inherent sparsity and complexity of clinical text pose substantial challenges to naive distance-based approaches.

6.2 MINHASH-BASED SIMILARITY DETECTION

A MinHash-based technique, coupled with *Locality Sensitive Hashing (LSH)*, was subsequently introduced to identify closely matching or duplicate discharge summaries. This approach was motivated by the suspicion that repeated clinical narratives could bias model training and inflate performance metrics without truly advancing model robustness. The text was preprocessed through *lowercasing, punctuation removal, and stopword filtration*, after which MinHash signatures were generated. These signatures were then indexed in an LSH structure to expedite the retrieval of pairs whose Jaccard similarity exceeded a user-defined threshold (e.g., 0.90).

As a cause, the repetition of routine sections within clinical notes (such as standardized instructions or procedural descriptions) led to the high similarity scores. *In effect*, these observations underscore the necessity for careful dataset curation, including the removal or consolidation of near-duplicate entries. Moreover, the identified pairs highlight potential avenues for validating the realism of *synthetic discharge notes*, since generated samples can be compared to authentic records through the same MinHash-based procedure.

6.3 KEY OBSERVATIONS AND CHALLENGES

- *Data Imbalance*: The imbalance in ICD codes—where common diagnoses overshadow rarer ones—was a major factor contributing to the low Macro F1-Score of the KNN baseline.

- *Complex Clinical Texts*: Clinical narratives often include technical jargon, abbreviations, and inconsistent structuring, which strains models that rely on straightforward vector distances.

- *Potential Data Redundancy*: The MinHash experiment exposed considerable text overlap, suggesting that a portion of the dataset may not add genuinely novel information for model training.

Taken together, these findings confirm that simplistic algorithms and untreated redundancies limit the performance and reliability of automated ICD coding. It is therefore anticipated that refined methods—especially those involving

transformer-based architectures and *knowledge-guided data augmentation*—will prove crucial for addressing rare codes, reducing the impact of repetitive content, and enhancing interpretability.

6.4 RARE DISEASE CO-OCCURRENCE PIPELINE

We developed an automated pipeline that extracts rare disease concepts from the ORDO ontology and maps them to SNOMED CT identifiers. Leveraging SNOMED CT concept and relationship files, a disease knowledge graph was built to capture clinically relevant associations between diseases. This graph was then traversed—using key relationships such as `is-a`, `associated with`, and `due to`—to identify groups of co-occurring conditions. Finally, SNOMED CT identifiers were translated to ICD-10-CM codes using a mapping file.

For instance, one generated group is:

Group 1: ['Q00.0', 'I48.91', 'J45.909', 'I25.10']

Here, the rare disease with code `Q00.0` corresponds to **Marfan syndrome**, while the common conditions are identified as **Atrial fibrillation** (`I48.91`), **Asthma** (`J45.909`), and **Coronary artery disease** (`I25.10`). Further work is ongoing on this to extract an an exhaustive set of groups of co-occurring diseases. The next phase report will contain all the final results and the next steps.

CHAPTER 7

CONCLUSION

A substantial foundation has been established through an extensive literature review, a clear problem definition, a detailed methodology, and a thorough understanding of the dataset. Emphasis has been placed on recognizing the complexity and breadth of ICD coding tasks, with particular attention to the underrepresentation of rare conditions. Although progress has been steady, the generation of synthetic data has not yet been executed, and it remains a pivotal step for enhancing the coverage and accuracy of ICD assignments.

Moving forward, the methodology's execution continues to be refined, with plans to employ knowledge-driven data augmentation to address class imbalance and to improve overall model performance. By integrating insights gained from preliminary investigations, a systematic approach to generating synthetic clinical narratives will be employed to bolster recognition of rare codes. The resulting models, once trained and evaluated, are expected to offer valuable contributions to automated ICD coding by delivering enhanced robustness and reliability.

7.1 GANTT CHART OF PLANNED WORK TIMELINE

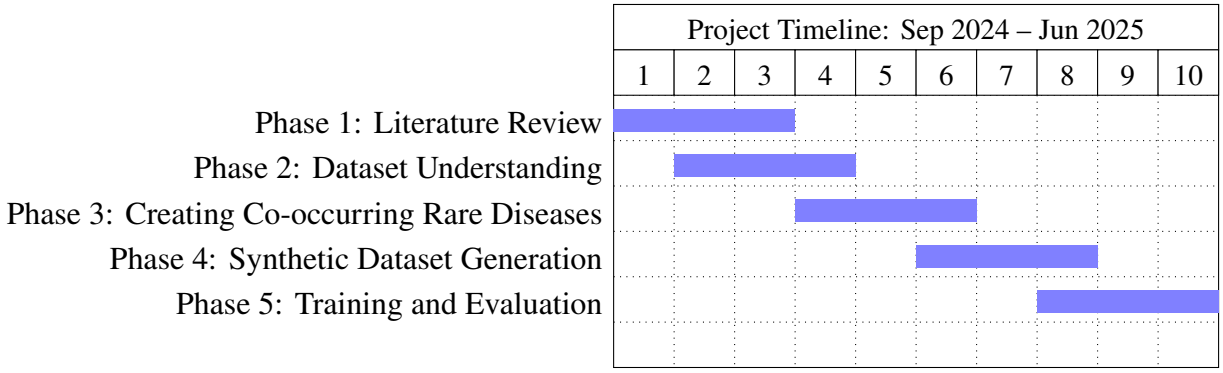

Figure 7.1: Gantt Chart Reflecting Main Phases and Overlaps

APPENDIX A

APPENDIX

A.1 SQL CODE

All SQL queries used throughout Chapter 5 are presented below. Each query is placed within a figure environment for clarity and reference.

```
-- Figure A.1: Counting rows/patients in the 'patients' table
SELECT
    COUNT(*) AS total_rows_in_patients,
    COUNT(DISTINCT subject_id) AS num_unique_patients
FROM mimic.patients;
```

Figure A.1: SQL code to retrieve the total number of rows and distinct patients in the `patients` table.

```
-- Figure A.2: Counting rows/admissions in the 'admissions' table
SELECT
    COUNT(*) AS total_rows_in_admissions,
    COUNT(DISTINCT hadm_id) AS num_unique_admissions
FROM mimic.admissions;
```

Figure A.2: SQL code to retrieve the total number of rows and distinct admissions in the `admissions` table.

```sql
-- Figure A.3: Identifying multiple admissions per patient
SELECT subject_id, COUNT(*) AS num_admissions
FROM mimic.admissions
GROUP BY subject_id
HAVING COUNT(*) > 1
ORDER BY num_admissions DESC
LIMIT 5;
```

Figure A.3: SQL code to find patients with more than one admission, illustrating recurrent hospital stays.

```sql
-- Figure A.4: Counting total discharge notes
SELECT COUNT(*) AS total_discharge_notes
FROM mimic.discharge;
```

Figure A.4: SQL code to count the total number of discharge summaries in the `discharge` table.

```sql
-- Figure A.5: Admissions with and without discharge notes
SELECT
    (SELECT COUNT(DISTINCT hadm_id) FROM mimic.admissions) AS total_admissions,
    (SELECT COUNT(DISTINCT hadm_id) FROM mimic.discharge)  AS hadm_with_discharge
    (SELECT COUNT(DISTINCT hadm_id) FROM mimic.admissions
     WHERE hadm_id NOT IN (
         SELECT DISTINCT hadm_id FROM mimic.discharge
     )
    ) AS hadm_without_discharge;
```

Figure A.5: SQL code comparing the number of admissions with discharge notes to those without.

```
-- Figure A.6: Word count distribution of discharge summaries
SELECT
    PERCENTILE_CONT(0.5)
        WITHIN GROUP (ORDER BY (LENGTH(text) - LENGTH(REPLACE(text, ' ', '')) + 1
      AS median_word_count,
    PERCENTILE_CONT(0.9)
      WITHIN GROUP (ORDER BY (LENGTH(text) - LENGTH(REPLACE(text, ' ', '')) + 1))
    AS p90_word_count,
    PERCENTILE_CONT(0.99)
    WITHIN GROUP (ORDER BY (LENGTH(text) - LENGTH(REPLACE(text, ' ', '')) + 1))
    AS p99_word_count
FROM mimic.discharge;
```

Figure A.6: SQL code for computing word-count percentiles of the `discharge` table.

```
-- Figure A.7: Comparing total code assignments in ICD-9 vs. ICD-10
SELECT
    icd_version,
    COUNT(*) AS code_assignments
FROM mimic.diagnoses_icd
GROUP BY icd_version
ORDER BY code_assignments DESC;
```

Figure A.7: SQL code to assess the distribution of ICD-9 and ICD-10 usage in the dataset.

```sql
-- Figure A.8: Retrieving top ICD-10 codes by frequency
SELECT icd_code, COUNT(*) AS freq
FROM mimic.diagnoses_icd
WHERE icd_version = 10
GROUP BY icd_code
ORDER BY freq DESC
LIMIT 10;
```

Figure A.8: SQL code used to find the 10 most common ICD-10 codes in MIMIC-IV.

```sql
-- Figure A.9: Counting unique ICD-10 codes
SELECT icd_version,
       COUNT(DISTINCT icd_code) AS num_unique_codes
FROM mimic.diagnoses_icd
GROUP BY icd_version;
```

Figure A.9: SQL code showing the total number of unique ICD codes by ICD version.

```sql
-- Figure A.10: Identifying ICD-10 codes with fewer than five occurrences
WITH code_counts AS (
    SELECT icd_code, COUNT(*) AS freq
    FROM mimic.diagnoses_icd
    WHERE icd_version = 10
    GROUP BY icd_code
)
SELECT COUNT(*) AS codes_under_5_occurrences
FROM code_counts
WHERE freq < 5;
```

Figure A.10: SQL code to quantify rare ICD-10 codes that appear fewer than five times.

```sql
-- Figure A.11: Calculating average ICD-10 codes per admission
WITH code_counts AS (
    SELECT hadm_id,
           COUNT(DISTINCT icd_code) AS code_count
    FROM mimic.diagnoses_icd
    WHERE icd_version = 10
    GROUP BY hadm_id
)
SELECT AVG(code_count) AS avg_icd10_codes_per_adm
FROM code_counts;
```

Figure A.11: SQL code to compute the average number of ICD-10 codes per admission.

```sql
-- Figure A.12: Merging discharge summaries with associated ICD-10 codes
DROP TABLE IF EXISTS mimic.full_notes_with_codes;

CREATE TABLE mimic.full_notes_with_codes AS
SELECT
    d.subject_id,
    d.hadm_id,
    d.text AS notes,
    STRING_AGG(di.icd_code, ',') AS icd_codes
FROM mimic.discharge AS d
JOIN mimic.admissions AS a
  ON d.subject_id = a.subject_id
    AND d.hadm_id = a.hadm_id
JOIN mimic.patients AS p
  ON p.subject_id = d.subject_id
JOIN mimic.diagnoses_icd AS di
  ON di.subject_id = d.subject_id
    AND di.hadm_id = d.hadm_id
WHERE di.icd_version = 10  -- focusing on ICD-10-CM only
GROUP BY
    d.subject_id,
    d.hadm_id,
    d.text;
```

Figure A.12: SQL code to create a consolidated table that pairs each discharge summary with its ICD-10 codes.

APPENDIX B

MINHASH PSEUDO-CODE

```
Algorithm 1: MinHash for Similar Document Detection


Input:
    - A collection of textual discharge summaries (D)
    - Number of permutations (num_perm)
    - Similarity threshold (t)


Output:
    - Set of document pairs whose Jaccard similarity exceeds t


Procedure:
1. Preprocess each document by:
   a) Converting text to lowercase
   b) Removing punctuation and non-alphanumeric characters
   c) Eliminating stopwords
2. Initialize an LSH index with threshold t and num_perm permutations.
3. For each document d in D:
   a) Create a MinHash object m with num_perm permutations
   b) For each token in d, update m using m.update(token.encode('utf8'))
   c) Insert the MinHash signature m into the LSH index
4. For each document d in D:
   a) Query the LSH index with d's signature to obtain similar documents
   b) Record document pairs that exceed the threshold
5. Return all identified pairs of documents with similarity above t
```

Figure B.1: Pseudocode for the MinHash Similarity Detection Process

APPENDIX C

APPENDIX: PSEUDOCODE

C.1 ONTOLOGY-GUIDED SYNTHETIC NOTE GENERATION

Figure C.1: Algorithm 1: Ontology-Guided Synthetic Note Generation

```
Input:
  r: Rare ICD code
  D: Disease cluster from ontology
  L: Language model with retrieval augmentation

Output:
  S: Synthetic discharge summary

Step 1:
  context_docs = RetrieveRelevantText(D, knowledge_source)
Step 2:
  structured_prompt = GenerateStructuredPrompt(D, context_docs)
  S_draft = L.generate(structured_prompt)
Step 3:
  for iteration in 1..MaxIter:
    feedback = CriticEvaluate(S_draft)
    if feedback indicates no major issues:
      break
    S_draft = L.refine(S_draft, feedback)
Step 4:
  if Validate(S_draft):
    S = S_draft
  else:
    S = None
```

C.2 HYBRID TRAINING WITH RARE-AWARE STRATEGIES

Figure C.2: Algorithm 2: Hybrid Training with Rare-Aware Strategies

```
Input:
  RealData = {(x_i, Y_i)}
  SyntheticData = {(x_j*, Y_j*)}
  Model parameters θ

Output:
  Trained model parameters θ

Step 1:
  TrainSet = RealData ∪ SyntheticData
  Initialize θ

Step 2:
  for epoch in 1..max_epochs:
    for batch B in WeightedSampleBatches(TrainSet):
      (X_B, Y_B) = PrepareBatch(B)
      if epoch >= epoch_focal_start:
        loss = FocalLoss(θ, X_B, Y_B) + BCE(θ, X_B, Y_B)
      else:
        loss = WeightedBCE(θ, X_B, Y_B)
      θ = θ - learning_rate * ∇_θ(loss)

Step 3:
  Evaluate on validation set and save best θ based on macro-F1.
```

BIBLIOGRAPHY

1. **Bengio, Y.**, **J. Louradour**, **R. Collobert**, and **J. Weston**, Curriculum learning. *In Proceedings of the 26th Annual International Conference on Machine Learning*. 2009.

2. **Bodenreider, O.** (2004). The unified medical language system (UMLS): integrating biomedical terminology. *Nucleic Acids Research*, **32**(Database issue), D267–D270.

3. **Campbell, S.** and **K. Giadresco** (2020). Computer-assisted clinical coding: A narrative review of the literature on its benefits, limitations, implementation, and impact on clinical coding professionals. *Health Information Management Journal*, **49**(1), 5–18.

4. **Cao, P.**, **Y. Chen**, **K. Liu**, **J. Zhao**, **S. Liu**, and **W. Chong**, Hypercore: Hyperbolic and co-graph representation for automatic ICD coding. *In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2020.

5. **Choi, E.**, **M. T. Bahadori**, **A. Schuetz**, **W. F. Stewart**, and **J. Sun** (2016). Doctor ai: Predicting clinical events via recurrent neural networks. *Journal of Machine Learning Research*, **17**(1), 359–396. URL `http://www.jmlr.org/papers/v17/15-322.html`.

6. **Devlin, J.**, **M.-W. Chang**, **K. Lee**, and **K. Toutanova**, Bert: Pre-training of deep bidirectional transformers for language understanding. *In Proceedings of NAACL-HLT*. 2019.

7. **Dong, H.**, **V. Suárez-Paniagua**, **W. Whiteley**, and **H. Wu** (2022). Automated clinical coding: what, why, and where we are? *npj Digital Medicine*, **5**, 159.

8. **Donnelly, K.** (2006). SNOMED CT: The advanced terminology and coding system for ehealth.

9. **Duque, E.**, **D. López**, and **M. Martínez** (2021). A knowledge-driven approach to automatic icd-10 coding in spanish clinical texts. *Journal of Biomedical Informatics*, **118**, 103749. URL `https://doi.org/10.1016/j.jbi.2021.103749`.

10. **Edin, J.**, **L. Borgholt**, **M. Maistro**, *et al.* (2024). An unsupervised approach to achieve supervised-level explainability in healthcare records. `https://arxiv.org/abs/2406.08958`. ArXiv preprint arXiv:2406.08958.

11. **Edin, J.**, **S. M. Jensen**, **M. Sahlgren**, **J. Lövgren**, and **A. Henriksson**, Automated medical coding on MIMIC-III and MIMIC-IV: a critical review and replicability study. *In Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2023.

12. **Excoffier, J.-B.**, **T. Roehr**, **A. Figueroa**, **K. Bressem**, and **M. Ortala** (2024).

Generalist embedding models are better at short-context clinical semantic search than specialized embedding models. https://arxiv.org/abs/2401.01943. ArXiv preprint arXiv:2401.01943.

13. **Falis, M.**, **H. Dong**, **A. Birch**, and **B. Alex**, Horses to zebras: Ontology-guided data augmentation and synthesis for ICD-9 coding. *In Proceedings of the 21st Workshop on Biomedical Language Processing (BioNLP)*. 2022.

14. **Falis, M.**, **A. P. Gema**, **H. Dong**, **L. Daines**, **S. Bassetti**, **M. H. Wong**, **R. S. Penfold**, **A. Birch**, and **B. Alex** (2024). Can GPT-3.5 generate and code discharge summaries? *Journal of the American Medical Informatics Association (JAMIA)*, **31**(10), 2284–2293.

15. **Farkas, R.** and **G. Szarvas** (2008). Automatic construction of rule-based icd-9-cm coding systems. *BMC Bioinformatics*, **9**(1).

16. **Gaebel, W.**, **J. Stricker**, and **A. Kerst** (2020). Changes from icd-10 to icd-11 and future directions in psychiatric classification. *Dialogues in Clinical Neuroscience*, **22**(1), 7–15.

17. **Gao, S.**, **M. Alawad**, **M. T. Young**, **J. Gounley**, **N. Schaefferkoetter**, **H. J. Yoon**, *et al.* (2021). Limitations of transformers on clinical text classification. *IEEE Journal of Biomedical and Health Informatics*, **25**(9), 3596–3607.

18. **Geiping, J.** and **T. Goldstein**, Cramming: Training a language model on a single gpu in one day. *In Proceedings of the International Conference on Machine Learning*. 2023.

19. **Gururangan, S.**, **A. Marasović**, **S. Swayamdipta**, **K. Lo**, **I. Beltagy**, **D. Downey**, and **N. A. Smith**, Don't stop pretraining: Adapt language models to domains and tasks. *In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. 2020. URL https://doi.org/10.18653/v1/2020.acl-main.750.

20. **Heo, T.-S.**, **B. Jo**, **Y. Yoo**, **K. Lee**, **Y. Park**, and **K. Kim**, Medical code prediction from discharge summary: Document to sequence bert using sequence attention. *In Proceedings of the 2022 International Conference on Electronics, Information, and Communication (ICEIC)*. 2022.

21. **Hochreiter, S.** and **J. Schmidhuber** (1997). Long short-term memory. *Neural Computation*, **9**(8), 1735–1780.

22. **Holzinger, A.**, **M. Dehmer**, and **I. Jurisica** (2017). What do we need to build explainable ai systems for the medical domain? https://arxiv.org/abs/1712.09923. ArXiv preprint arXiv:1712.09923.

23. **Huang, C.-W.**, **S.-C. Tsai**, and **Y.-N. Chen**, Plm-icd: Automatic icd coding with pre-trained language models. *In Proceedings of the 4th Clinical Natural Language Processing Workshop*. 2022.

24. **Ji, S.**, **P. Hölttä**, and **P. Marttinen** (2021). Does the magic of bert apply to medical code

assignment? a quantitative study. *Artificial Intelligence in Medicine*, **117**, 102081.

25. **Johnson, A. E. W.**, **L. Bulgarelli**, **L. Shen**, **A. Gayles**, **A. Shammout**, **S. Horng**, **T. J. Pollard**, **S. Hao**, **B. Moody**, **B. Gow**, **L. H. Lehman**, **L. A. Celi**, and **R. G. Mark** (2023). Mimic-iv, a freely accessible electronic health record dataset. *Scientific Data*, **10**(1), 1.

26. **Johnson, A. E. W.**, **T. J. Pollard**, **L. Shen**, *et al.* (2016). Mimic-iii, a freely accessible critical care database. *Scientific Data*, **3**(160035).

27. **Kim, Y.**, Convolutional neural networks for sentence classification. *In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2014.

28. **Kraljevic, Z.**, **T. Searle**, **A. Shek**, *et al.* (2021). Multi-domain clinical natural language processing with medcat: The medical concept annotation toolkit. *Artificial Intelligence in Medicine*, **117**, 102083.

29. **Kumichev, G.**, **P. Blinov**, **Y. Kuzkina**, *et al.* (2024). Medsyn: Llm-based synthetic medical text generation framework. `https://arxiv.org/abs/2408.02056`. ArXiv preprint arXiv:2408.02056.

30. **LeCun, Y.**, **Y. Bengio**, and **G. Hinton** (2015). Deep learning. *Nature*, **521**(7553), 436–444.

31. **Li, F.** and **H. Yu**, Icd coding from clinical text using multi-filter residual convolutional neural network. *In Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34. 2020.

32. **Mullenbach, J.**, **S. Wiegreffe**, **J. Duke**, **J. Sun**, and **J. Eisenstein**, Explainable prediction of medical codes from clinical text. *In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics*. 2018.

33. **Nguyen, T.-T.**, **V. Schlegel**, **A. Kashyap**, **S. Winkler**, **S.-S. Huang**, **J.-J. Liu**, and **C.-J. Lin**, Mimic-iv-icd: A new benchmark for extreme multi-label classification. *In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*. 2023.

34. **Nickel, M.** and **D. Kiela**, Poincaré embeddings for learning hierarchical representations. *In Advances in Neural Information Processing Systems*. 2017.

35. **Nussbaum, Z.**, **J. X. Morris**, **B. Duderstadt**, and **A. Mulyar** (2024). Nomicbert: Training a reproducible long context text embedder. `https://arxiv.org/abs/2402.01613`. ArXiv preprint arXiv:2402.01613.

36. **Organization, W. H.** (2019). International classification of diseases 11th revision (icd-11).

37. **Perotte, A.**, **F. Wood**, **A. Simma**, **M. Braverman**, **N. Elhadad**, and **S. Ghosh** (2014). Diagnosis code assignment: models and evaluation metrics. *Journal of the American Medical Informatics Association*, **21**(2), 231–237.

38. **Pollard, T. J.**, **A. E. Johnson**, **J. D. Raffa**, **L. A. Celi**, **R. G. Mark**, **B. Moody**, *et al.* (2018). The eicu collaborative research database, a freely available multi-center database for critical care research. *Scientific Data*, **5**, 180178. URL `https://doi.org/10.1038/sdata.2018.178`.

39. **Portes, J.**, **A. Trott**, **S. Havens**, *et al.*, Mosaicbert: A bidirectional encoder optimized for fast pretraining. *In Advances in Neural Information Processing Systems*, volume 36. 2023.

40. **Ren, W.**, **R. Zeng**, **T. Wu**, **T. Zhu**, and **R. G. Krishnan**, Hicu: Leveraging hierarchy for curriculum learning in automated icd coding. *In Proceedings of the Machine Learning for Healthcare Conference*. 2022.

41. **Rios, A.** and **R. Kavuluru**, Few-shot and zero-shot multi-label learning for structured label spaces. *In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 2018.

42. **Savova, G. K.**, **J. J. Masanz**, **P. V. Ogren**, **J. Zheng**, **S. Sohn**, **K. C. Kipper-Schuler**, and **C. G. Chute** (2010). Mayo clinical text analysis and knowledge extraction system (cTAKES): architecture, component evaluation and applications. *Journal of the American Medical Informatics Association*, **17**(5), 507–513.

43. **Scheurwegs, E.**, **K. Luyckx**, **L. Luyten**, and **W. Daelemans** (2017). Data integration of structured and unstructured sources for assigning clinical codes to patient stays. *Journal of the American Medical Informatics Association*, **24**(e1), e68–e76.

44. **Teng, F.**, **W. Yang**, **L. Chen**, **L. Huang**, and **Q. Xu** (2020). Explainable prediction of medical codes with knowledge graphs. *Frontiers in Bioengineering and Biotechnology*, **8**, 867.

45. **Vu, T.**, **D. Q. Nguyen**, and **A. Nguyen** (2020*a*). A label attention model for icd coding from clinical text. *Journal of Biomedical Informatics*, **102**, 103381.

46. **Vu, X. Y. H.**, **A. T. Luu**, **M. Le**, **T. N. Nguyen**, **T. Atasoy**, and **O. Bousquet**, A label attention model for icd coding from clinical text. *In Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*. 2020*b*.

47. **Warner, B.**, **A. Chaffin**, **B. Clavié**, *et al.* (2024). Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. `https://arxiv.org/abs/2309.16609`. ArXiv preprint arXiv:2309.16609.

48. **Wrenn, J. O.**, **D. M. Stein**, **S. Bakken**, and **P. D. Stetson** (2010). Quantifying clinical narrative redundancy in an electronic health record. *Journal of the American Medical*

Informatics Association, **17**, 49–53.

49. **Wu, H.**, **J. Fan**, **J. Li**, **J. Massoudi**, **D. Cushen**, **R. J. B. Dobson**, and **F. C. G. Taylor** (2018). SemEHR: A general-purpose semantic search system to surface semantic data from clinical notes for tailored care, trial recruitment, and clinical research. *Journal of the American Medical Informatics Association*, **25**(5), 530–537.

50. **Xie, X.**, **Y. Xiong**, **P. S. Yu**, and **Y. Zhu**, Ehr coding with multi-scale feature attention and structured knowledge graph propagation. *In Proceedings of the 28th ACM International Conference on Information and Knowledge Management*. 2019.

51. **Yang, Z.**, **S. Kwon**, **Z. Yao**, and **H. Yu**, Multi-label few-shot ICD coding as autoregressive generation with prompt. *In Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI)*. 2023.

52. **Yuan, Z.**, **C. Tan**, and **S. Huang**, Code synonyms do matter: Multiple synonyms matching network for automatic ICD coding. *In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL) – Volume 2: Short Papers*. 2022.

53. **Zhang, X.**, **Y. Zhang**, **D. Long**, and **W. Xie**, Gte-en-mlm: Generalized long-context text representation for retrieval tasks. *In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 2024.